

Chapter 1: Principles of Survey Sampling

Sampling is the process of selecting a subset of units from a target population in order to estimate characteristics of the whole population. In official statistics, the sampling frame is the list of all units from which the sample is drawn. An incomplete or outdated sampling frame introduces coverage error, which cannot be corrected by increasing the sample size.

Simple random sampling gives every unit in the population an equal and independent probability of selection. Stratified sampling first divides the population into homogeneous strata and then draws an independent sample from each stratum. Stratification reduces sampling variance when the strata are internally homogeneous with respect to the study variable.

Systematic sampling selects every k -th unit after a random start, where k is the sampling interval defined as N divided by n . Systematic sampling is operationally simple in field work but can produce biased estimates if the frame has a periodicity that coincides with the sampling interval.

Chapter 2: Survey Design and Questionnaire Construction

The reference period of a survey is the time interval to which the reported information relates. A shorter reference period reduces recall error but increases the sampling variance of rare events. The Periodic Labour Force Survey uses both a seven-day current weekly status and a 365-day usual status reference period.

Question wording must avoid double-barrelled questions, which ask about two distinct issues in a single item. Leading questions that suggest a preferred answer introduce response bias. Pre-testing the questionnaire in a pilot survey is a mandatory step before full-scale field deployment.

The design effect, denoted $deff$, is the ratio of the variance of an estimate under the actual complex sample design to the variance that would have been obtained under simple random sampling of the same size. A design effect greater than one indicates a loss of precision relative to simple random sampling.

Chapter 3: Data Collection and Non-Response

Non-response occurs when a selected unit does not provide the required information. Unit non-response refers to a complete failure to obtain data from a sampled unit, while item non-response refers to missing answers for particular questions within an otherwise completed schedule.

Non-response bias arises when non-respondents differ systematically from respondents with respect to the characteristic being measured. Increasing the sample size does not reduce non-response bias; it only reduces sampling variance. Weighting adjustments and repeated call-backs are the standard mitigation measures.

Computer Assisted Personal Interviewing, or CAPI, allows validation rules to be applied at the point of data capture. Built-in range checks and consistency checks reject impossible values such as an age of 250 years at the moment of entry, reducing the cost of downstream editing.

Chapter 4: Data Quality Frameworks

Data quality in official statistics is assessed along several dimensions: relevance, accuracy, timeliness, punctuality, accessibility, comparability and coherence. Accuracy is further decomposed into sampling error and non-sampling error.

A completeness check verifies that all expected records and all mandatory fields are present. A uniqueness check detects duplicate records that would otherwise inflate estimates. A plausibility check tests whether a reported value lies within a credible range for that variable.

Metadata documents the definitions, classifications, reference period, coverage and methodology of a dataset. Without adequate metadata, users cannot correctly interpret a published statistic, and the dataset cannot be reliably compared across time periods or with other sources.